AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 16 · WORKBOOK

Verify Numbers and Dates

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected numeric and date verification record.

Explain the checks, trace the percentage, reject false certainty and design a revealing case.

Keep one habit: Separate new responses from preserved citation review; keep unresolved dates explicit.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Multimodal **Evidence**. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Image Generation Models. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Source Quality

Multimodal Evidence

Image Generation Models

W1 · Explain the separate checks

> Current tasks have separate fields; earlier citation answers remain preserved.

Complete main record

Main fictional record: 15/20 trials passed; draft says 70% and meeting June 31. June has 30 days; no intended meeting date is supplied. Changed case: 14/20 passes, with no new date evidence. Use source identity, numerator/denominator, units and calendar validity as separate checks.

Current task and response

- Explain numerator, denominator and percentage using the main record. Describe why source identity, calendar validity and intended event date need separate checks.

> **My numeric and calendar checks**

▼ **Review · Earlier citation task and saved response**

Reference for this exercise: open this week's student lesson, "Verify Numbers and Dates", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Recalculate and preserve uncertainty

> Current tasks have separate fields; earlier citation answers remain preserved.

Complete main record

Main fictional record: 15/20 trials passed; draft says 70% and meeting June 31. June has 30 days; no intended meeting date is supplied. Changed case: 14/20 passes, with no new date evidence. Use source identity, numerator/denominator, units and calendar validity as separate checks.

Current task and response

- Use the fictional record of 15 passes out of 20 trials and the draft 70%, meeting June 31. Show the fraction and percentage, state the supplied calendar fact, and separate the computable correction from the missing intended date. Then change only the count to 14 passes out of 20 and explain what changes and what remains unresolved.

My numeric and calendar checks

▼ **Review · Earlier citation task and saved response**

An answer has four factual claims. Three have citations, but only two are supported by their cited passages. All links exist. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Reject an unsupported replacement

> Current tasks have separate fields; earlier citation answers remain preserved.

Complete main record

Main fictional record: 15/20 trials passed; draft says 70% and meeting June 31. June has 30 days; no intended meeting date is supplied. Changed case: 14/20 passes, with no new date evidence. Use source identity, numerator/denominator, units and calendar validity as separate checks.

Current task and response

- A calculator confirms 15/20 = 75%, and a calendar rejects June 31. Does this establish the real trial reliability and the intended meeting date? Explain the two remaining evidence gaps.

My numeric and calendar checks

▼ **Review · Earlier citation task and saved response**

A report omits this qualification: Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Design a revealing verification case

> Current tasks have separate fields; earlier citation answers remain preserved.

Complete main record

Main fictional record: 15/20 trials passed; draft says 70% and meeting June 31. June has 30 days; no intended meeting date is supplied. Changed case: 14/20 passes, with no new date evidence. Use source identity, numerator/denominator, units and calendar validity as separate checks.

Current task and response

- Design one complete fictional variant testing a wrong denominator, incompatible units or a valid-looking unsupported date. Supply all facts needed for your expected answer; record missing intent explicitly. Predict the result before running a tool. Keep the independent learner case unused.

My numeric and calendar checks

▼ Review · Earlier citation task and saved response

Reference for this exercise: open this week's student lesson, "Verify Numbers and Dates", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

W4 · Audit a fresh learner percentage and deadline independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Use only the supplied learner counts, claimed percentage and deadline. Show your checks and identify any additional source needed. No actual attendance data or approved deadline notice is supplied.

A draft claims18 of24 learners is**60%**, and gives an April31 deadline. Check both statements without inventing a deadline.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Route arithmetic to a calculator and dates to direct sources before presenting them.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Earlier citation completion practice: retain this task and saved response as review. Use current W2 for the supported numeric and calendar practice.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Claim support and citation evaluation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A report says 21/28=**70%** and gives February 30 as the deadline. Correct what can be computed and identify what must be checked externally.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

An answer says 15/20=**80%** and sets a deadline of February 30.

Verify both.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A report says 12 of 40 responses is 40%. **Recalculate** and state the correction with the denominator.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 16 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A report says 21/28=**70%** and gives February 30 as the deadline. Correct what can be computed and identify what must be checked externally.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Additional practice · Earlier citation-count task

REFERENCE · SECTION 15

Additional practice · Earlier citation-count task

> This retained exercise differs from the selected numeric/date assessment.

Earlier task and saved response

- Five claims; four cited; three supported by cited passages; one cited ID is nonexistent. Compute support and citation presence and explain link limits.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

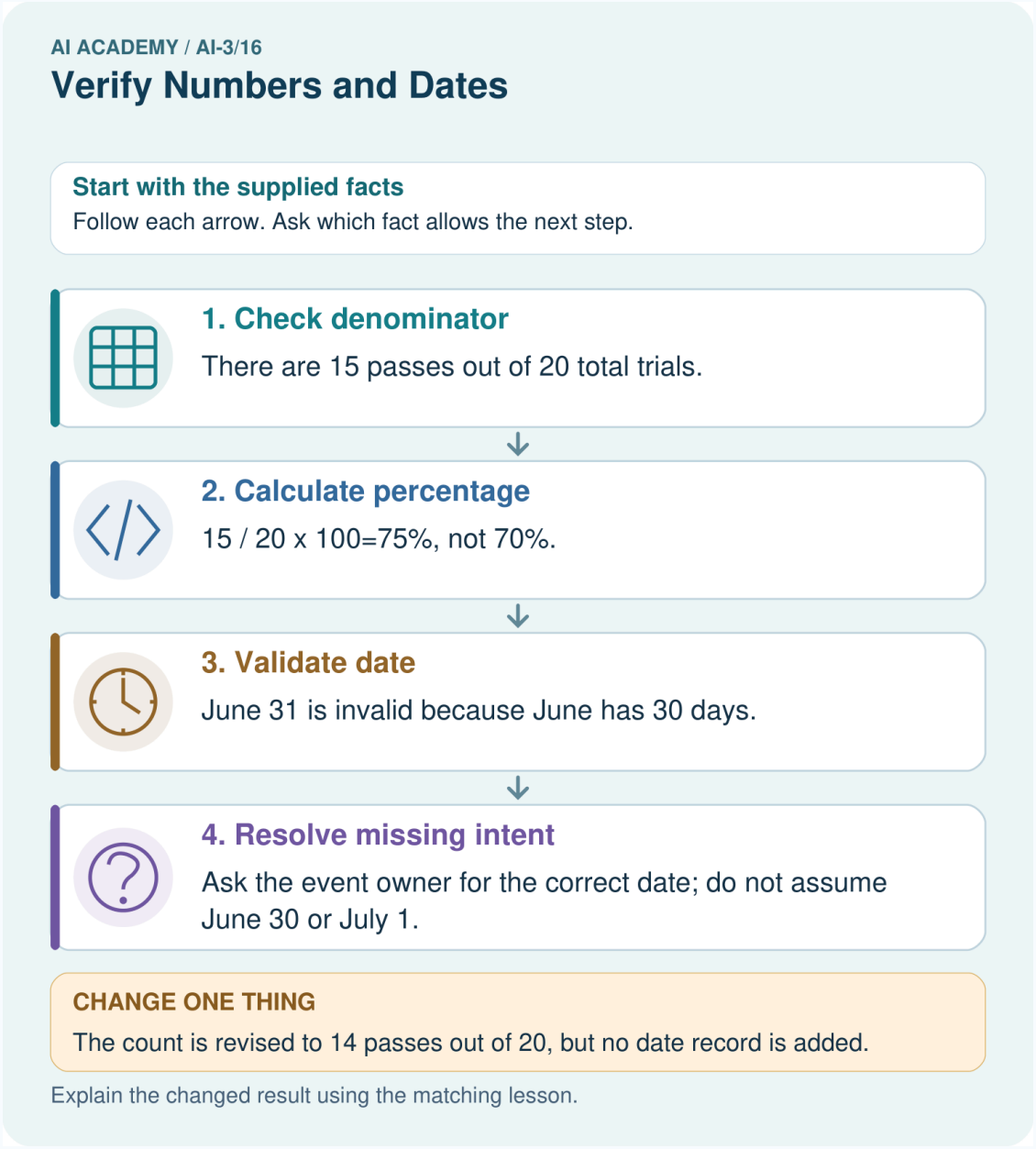
> Keep the explanation and the support you used.

Use this when helpful

A fictional Study Buddy checks a science-club summary against supplied counts and calendar facts.

Record: 15 of 20 fictional model trials passed. Draft: "**70%** passed; the meeting is on June 31."

June has 30 days. No authoritative record gives the intended meeting date.



Follow the labelled picture with the matching explanation. Open the vector diagram

Which part can be corrected immediately from existing **evidence**, and which needs follow-up?

> **My picture-based explanation and changed-case prediction**

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Make Study Buddy check quantities, **unit**s, dates and source support separately; record unresolved fields instead of guessing.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Make Study Buddy check quantities, **unit**s, dates and source support separately; record unresolved fields instead of guessing.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional Study Buddy checks a science-club summary against supplied counts and calendar facts.
- Which labelled step matches this detail? Ask the event owner for the correct date; do not assume June 30 or July 1.
- 1. Validate date

- 2. Check denominator
- 3. Resolve missing intent
- 4. Calculate percentage
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Resolve missing intent Ask the event owner for the correct date; do not assume June 30 or July 1. A percentage requires the correct numerator and denominator.

Calendar validation identifies an impossible date, but it cannot discover what date the author intended.

These checks use different **evidence**: arithmetic repairs the percentage from the supplied counts, while the date needs another authoritative source before publication.

Changed-case question

- Changed-case practice: The count is revised to 14 passes out of 20, but no date record is added. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. **70%** now matches the count; June 31 remains invalid and unresolved.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: **70%** now matches the count; June 31 remains invalid and unresolved. 14÷20×100=**70%**; changing a count provides no new calendar **evidence**.

Repair and save your work

Review Source Quality (ai-3/15). Goal: Explain the **mechanism** and terminology of conflicting and time-sensitive sources. Short practice: Which source fields must be checked before answering day 9?.

Return to this week and explain what you changed.

Review Multimodal **Evidence** (ai-3/13). Goal: Keep visible **evidence**, hidden contents and caption claims separate. Short practice: Why were both “exactly three” and “exactly five” unsupported before opening the box?. Return to this week and explain what you changed.

Review Image Generation Models (ai-3/10). Goal: Explain the **mechanism** and terminology of image generation and provenance.

Short practice: What may these two grids explain, and what can they not establish about a real generator?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

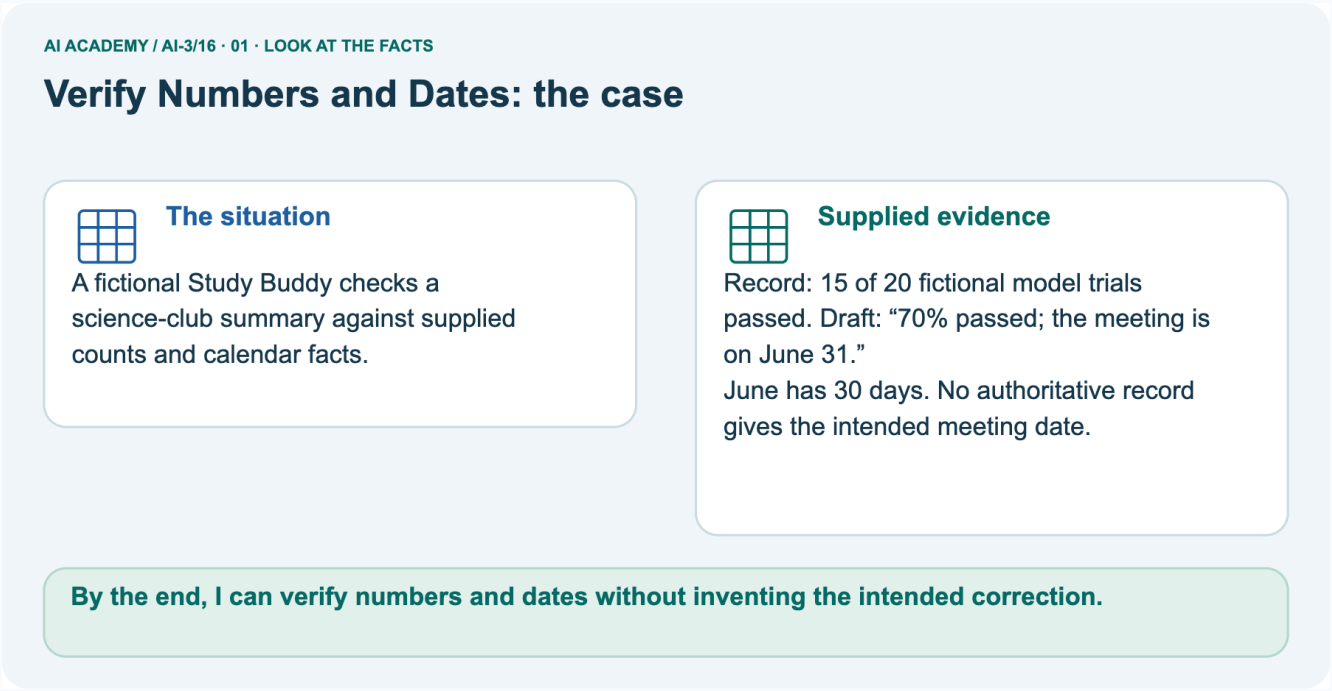
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

Choose the image that helps rather than repeating all six.

01. Verify Numbers and Dates: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

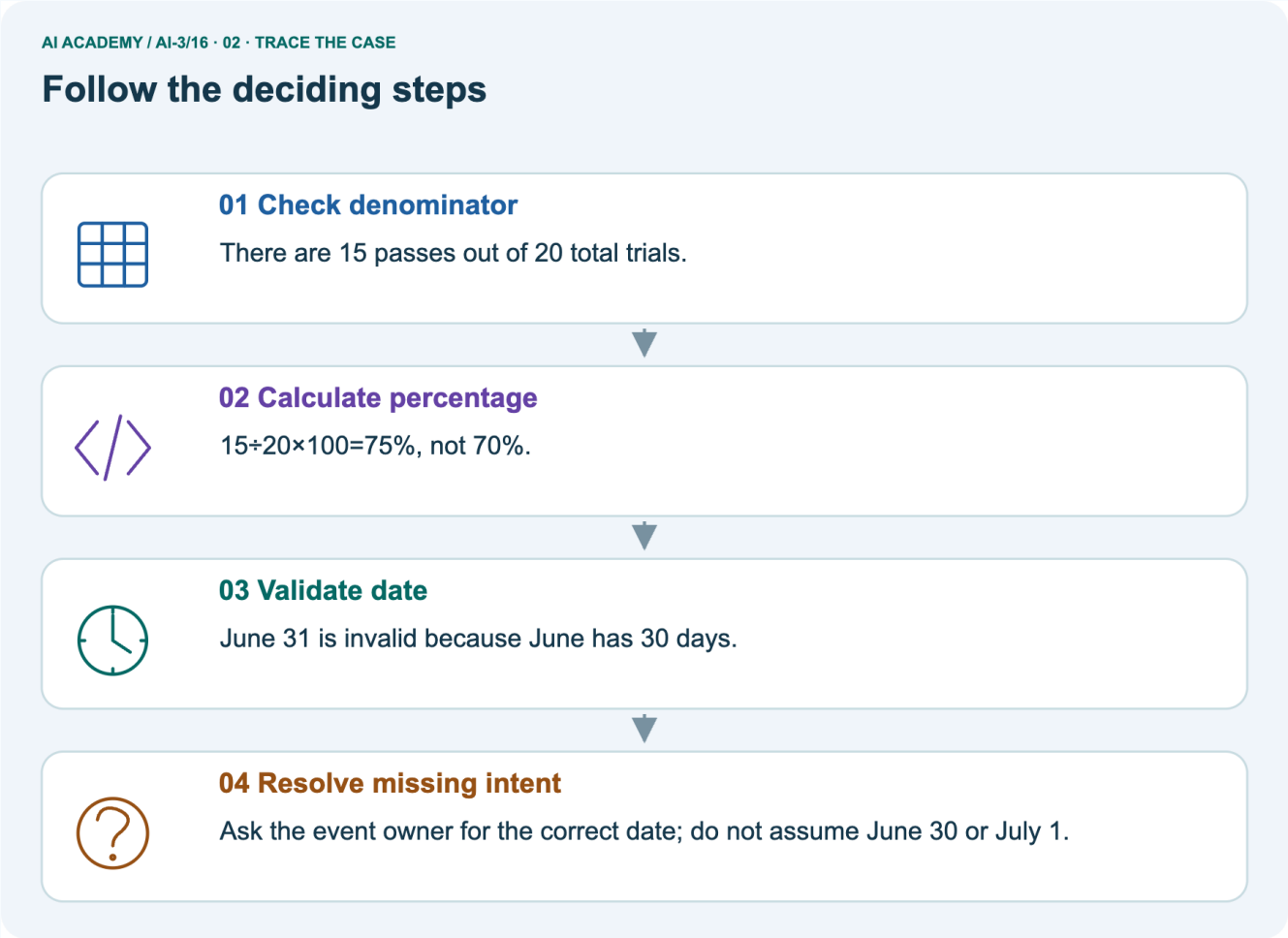
A fictional Study Buddy checks a science-club summary against supplied counts and calendar facts. Supplied facts: Record: 15 of 20 fictional model trials passed.

Draft: “**70%** passed; the meeting is on June 31.” June has 30 days. No authoritative record gives the intended meeting date.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



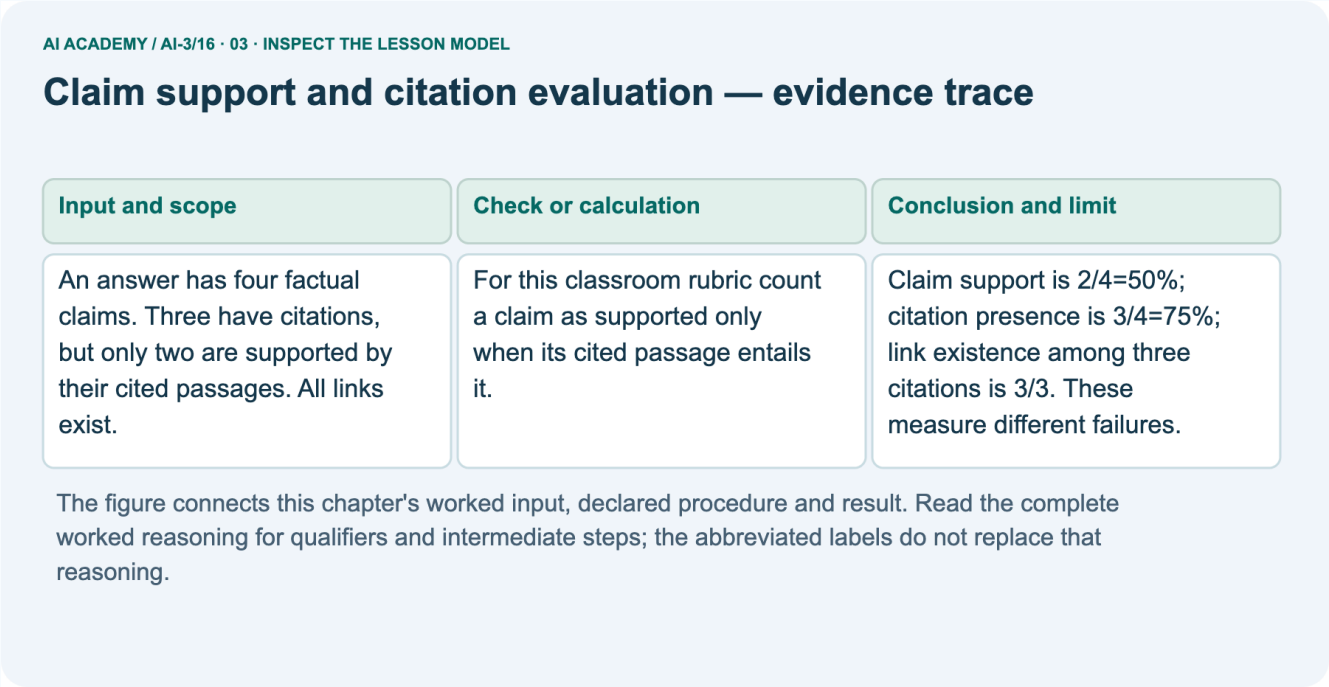
Follow the labelled picture with the matching explanation. Open the vector diagram

Check denominator: There are 15 passes out of 20 total trials. → Calculate percentage: 15÷20×100=75%, not **70%**. → Validate date: June 31 is invalid because June has 30 days. → Resolve missing intent: Ask the event owner for the correct date; do not assume June 30 or July 1.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Claim support and citation evaluation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Claim support and citation evaluation — **evidence** trace. An answer has four factual claims. Three have citations, but only two are supported by their cited passages.

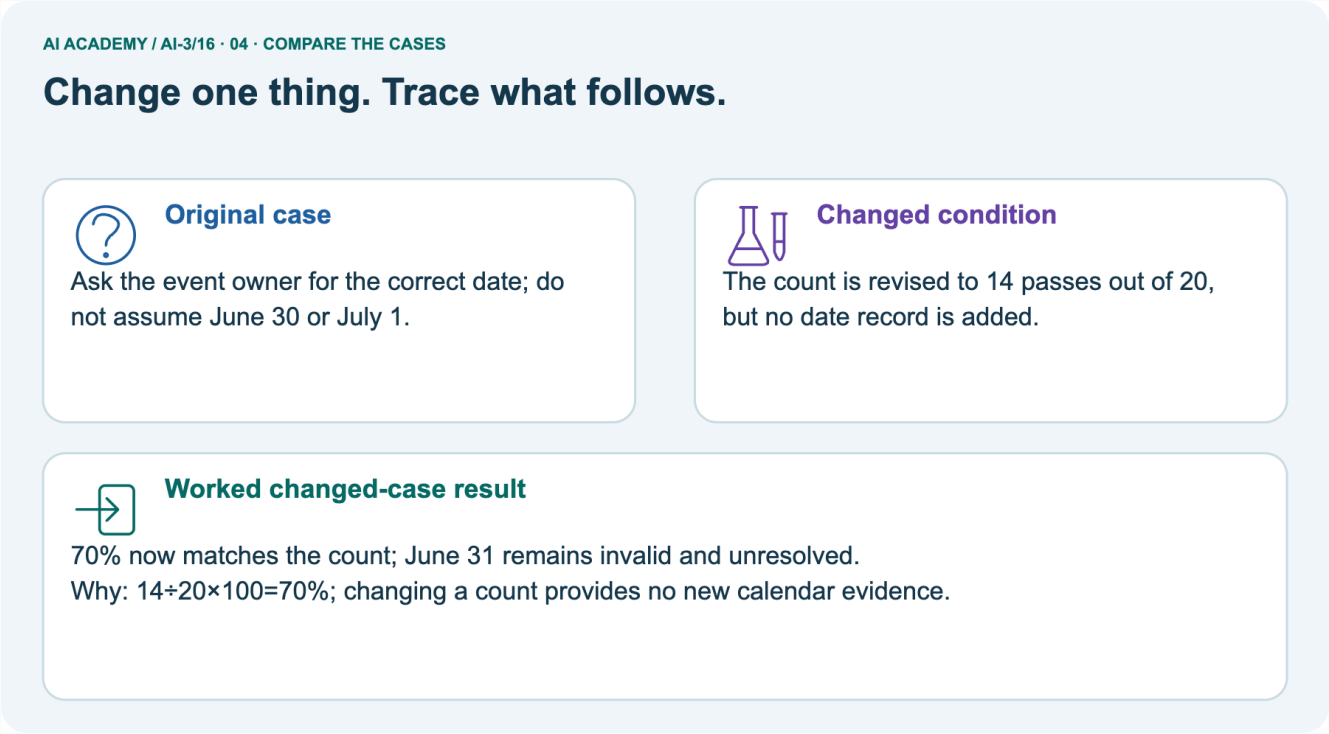
All links exist. | For this classroom rubric count a claim as supported only when its cited passage entails it. | Claim support is 2/4=50%; citation presence is 3/4=75%; link existence among three citations is 3/3.

These measure different failures.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

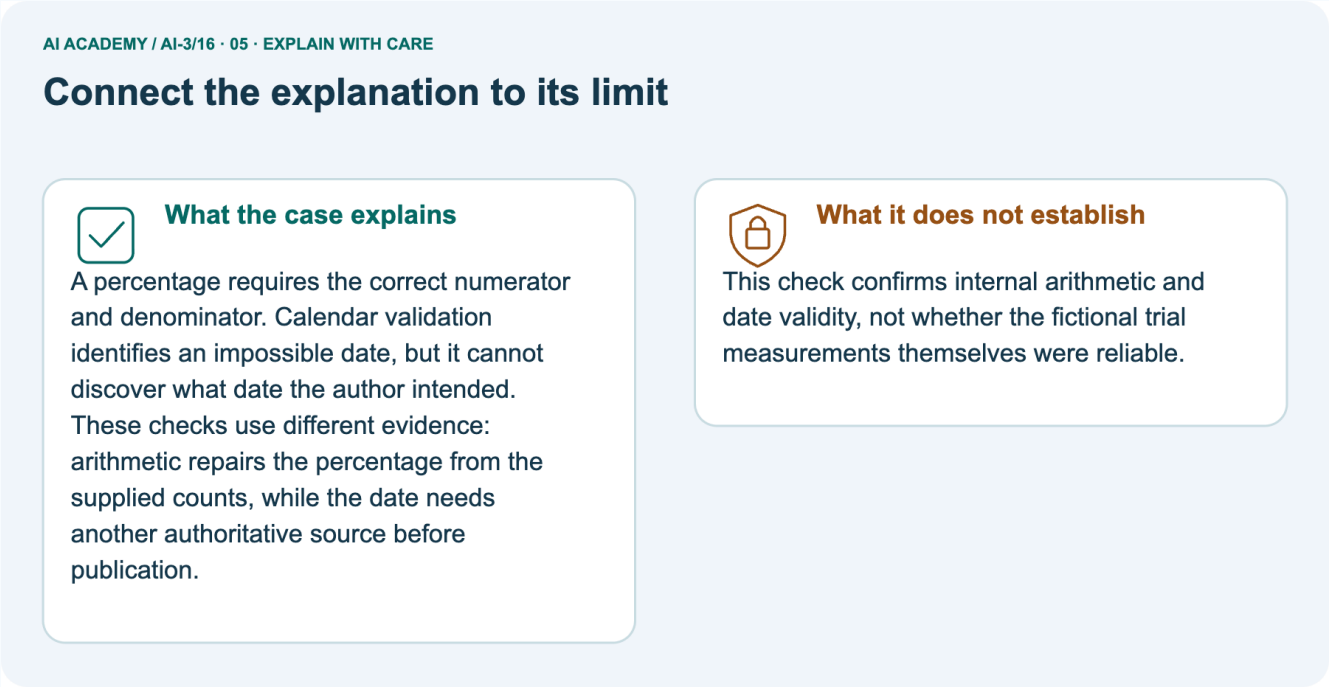
Original case: Ask the event owner for the correct date; do not assume June 30 or July 1.
Changed condition: The count is revised to 14 passes out of 20, but no date record is added.

Expected result: **70%** now matches the count; June 31 remains invalid and unresolved.
Reason: 14÷20×100=**70%**; changing a count provides no new calendar **evidence**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A percentage requires the correct numerator and denominator. Calendar validation identifies an impossible date, but it cannot discover what date the author intended.

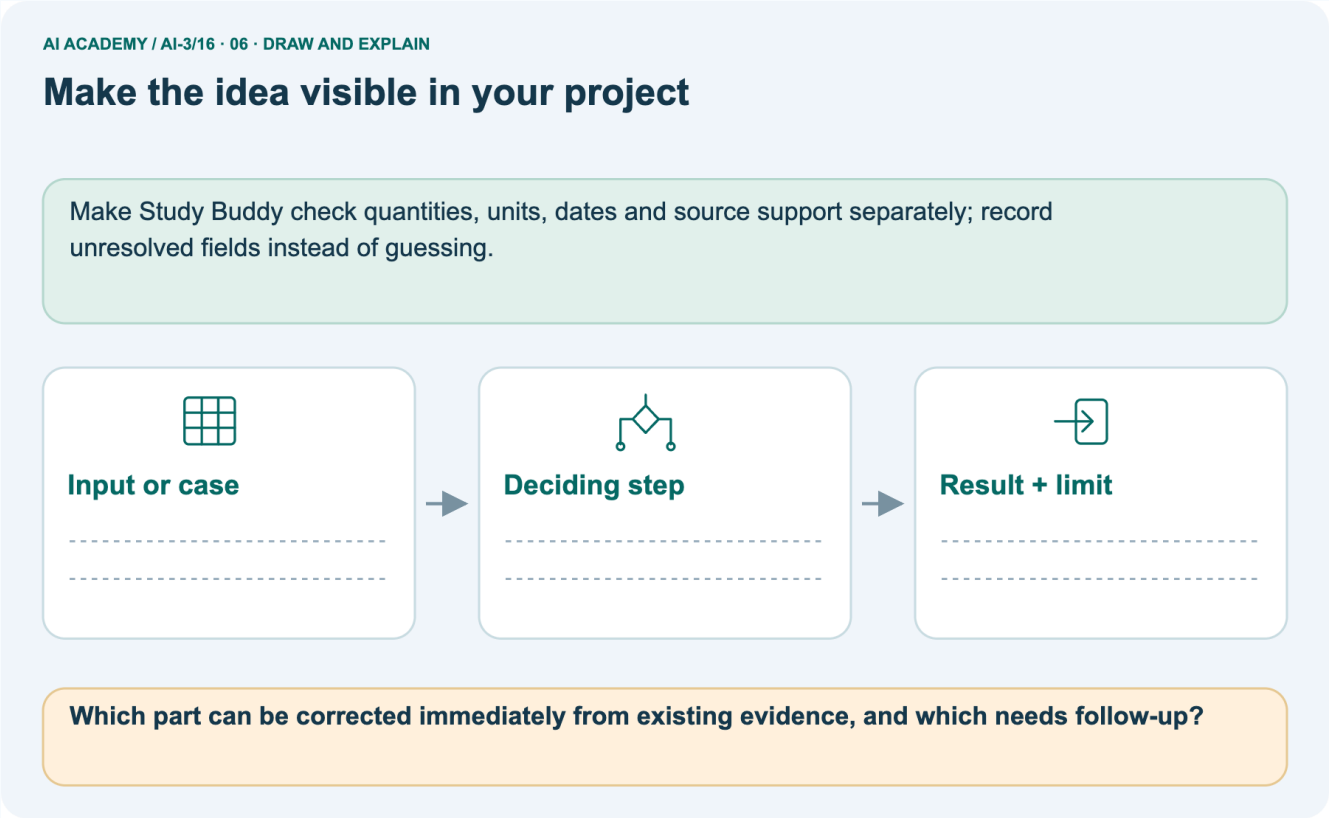
These checks use different **evidence**: arithmetic repairs the percentage from the supplied counts, while the date needs another authoritative source before publication.

Limit: This check confirms internal arithmetic and date validity, not whether the fictional trial measurements themselves were reliable.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Make Study Buddy check quantities, **unit**s, dates and source support separately; record unresolved fields instead of guessing. Draw your **input**, deciding step, result and limit.

Supported question: Which part can be corrected immediately from existing **evidence**, and which needs follow-up?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Earlier wording, alternatives and picture transcripts

01. Verify Numbers and Dates: the case

02. Follow the deciding steps

03. Claim support and citation evaluation — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.